29/9/25

100 prisoners:

$$p(\text{all win}) = p\left(\begin{matrix}\text{random permutation has} \\ \text{no cycle longer than } 50\end{matrix}\right)$$

Compute # permutation with a length $l > 50$

For $l \geq 50$, only 1 cycle is possible

$$\binom{100}{l} \cdot (l-1)! \cdot (100-l)! = \frac{100!}{l!\,\cancel{(100-l)!}} \cdot (l-1)! \cdot \cancel{(100-l)!} = \frac{100!}{l \cdot \cancel{(l-1)!}} \cdot \cancel{(l-1)!} = \frac{100!}{l}$$

$\binom{100}{l}$: elements of the cycle

$(l-1)!$: their order

$$\sum_{l=51}^{100} \frac{100!}{l} = 100! \sum_{l=51}^{100} \frac{1}{l} = 100! \cdot (H_{100} - H_{50})$$

H_{100}: nth harmonic number

$$p(\text{close}) = 100! \cdot (H_{100} - H_{50}) \cdot \frac{1}{100!}$$

$$p(\text{win}) = 1 - (H_{100} - H_{50}) \approx 0.3$$

MISSING NUMBER → acc ^ i ^ arr[i]

Duplicate elements in array:

array $A[0, \dots, n]$ of $n+1$ numbers in $\{0, \dots, n-1\}$

By pidgeonholes principles, there must be atleast a duplicate elements

Direct access Table → Binary vector B $\boxed{\;|\qquad\qquad|\;}$ (0 … n)

$B[i] = 1$ insert

$B[i] = 0$ remove

$\Theta(n)$ – $\Theta(n)$

Time — extraspace

A $\boxed{1\ 3\ 5\ 6\ 5\ 2\ 0\ 4\ 7}$

#1	0	0	1	1	1	0	0	1	1
#2	0	1	0	1	0	1	0	0	1
#3	1	1	1	0	1	0	0	0	1

$e = \boxed{1|0|1} = 5$

1) $\begin{array}{c|c} 0 & 1 \\ \hline 4 & \underline{5} \end{array}$

2) $\begin{array}{c|c} 0 & 1 \\ \hline 2 & 2 \end{array}$ restrcit to the number starting with 1

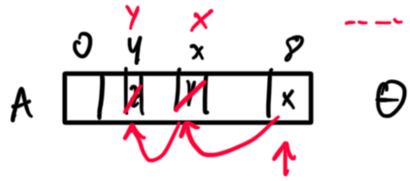
$\Theta(n \log n) - \Theta(1)$

time space

3) $\begin{array}{c|c} 0 & 1 \\ \hline 1 & 2 \end{array}$ restart to withto

Best if No Access => Does Random access help?

Destroy A



$\Theta(n) - O(1)$ but destroying A

without mark

at time 0 F, S are in the initial position (arr[n])

F, S pointers $\rightarrow$ F = 2 step

S = 1 step